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Exhaust purification apparatus

Abstract

An exhaust purification apparatus capable of improving stirring property of urea water and properly cooling an injection device is provided. The exhaust purification apparatus that purifies nitrogen oxides in exhaust gas includes an exhaust pipe that causes exhaust gas discharged from an engine to flow, a catalyst reactor (catalyst) provided in the exhaust pipe to reduce the nitrogen oxides in the exhaust gas, a through pipe provided in such a way as to penetrate the exhaust pipe, and an injection device provided inside the through pipe to inject a reducing agent for reducing exhaust gas into the exhaust pipe.

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Background/Summary

(1) This application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2023-195630, filed on Nov. 17, 2023, which is incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to an exhaust purification apparatus that purifies nitrogen oxides in exhaust gas.

BACKGROUND ART

(3) Conventionally, an exhaust purification apparatus purifies exhaust gas discharged from an engine by reducing nitrogen oxides contained in the exhaust gas, and is constituted by a so-called SCR system. The exhaust purification apparatus is configured to reduce nitrogen oxides accumulated in a catalyst by providing the catalyst in an exhaust passage of the exhaust gas and supplying urea into the exhaust gas.

(4) Further, for example, in an exhaust purification apparatus disclosed in Patent Document 1, a urea water injection nozzle is provided on a downstream side of an exhaust changeover valve in an exhaust pipe. A double pipe of the urea water injection nozzle is arranged in such a way that the downstream side is positioned inside the exhaust pipe and an upstream side is positioned outside the exhaust pipe, and an air nozzle of the urea water injection nozzle is arranged on a downstream side of a liquid nozzle in such a way that an upstream side end thereof abuts on a downstream side end part of the double pipe. Namely, the exhaust purification apparatus is provided with a urea water injection nozzle that protrudes from a wall of the exhaust pipe toward the central part.

PRIOR ART DOCUMENT

Patent Document

(5) Patent Document 1: Japanese Unexamined Patent Publication No. 2014-234793

SUMMARY OF INVENTION

Technical Problem

(6) In the exhaust purification apparatus, it is necessary to cool an injection device such as an injection valve or an injection nozzle for injecting urea water in order to suppress thermal damage to the urea water due to exposure of the injection device to the exhaust gas. In the urea water injection nozzle disclosed in Patent Document 1, a cooling method of cooling the nozzle by using air used for injection of the urea water is applied. However, the injection device may not be appropriately cooled depending on a configuration of the exhaust purification apparatus. For example, in an exhaust purification apparatus to which an airless injection device that does not use air for injection of urea water is applied, since the injection device cannot be cooled by air, an other cooling means needs to be provided. Note that, depending on the arrangement of the injection device or the arrangement of the cooling means in the exhaust pipe, there is a possibility that stirring property of the exhaust gas and the urea water may be reduced.

(7) An object of the present invention is to provide an exhaust purification apparatus capable of improving the stirring property of urea water and appropriately cooling an injection device.

Solution to Problem

(8) In order to solve the above-described problem, an exhaust purification apparatus according to an aspect of the present invention, which purifies nitrogen oxides in exhaust gas, includes: an exhaust pipe that causes the exhaust gas discharged from an engine to flow; a catalyst that is provided in the exhaust pipe and that reduces the nitrogen oxides in the exhaust gas; a through pipe that is provided in such a way as to penetrate the exhaust pipe; and an injection device that is provided inside the through pipe and that injects a reducing agent for reducing the exhaust gas into the exhaust pipe.

Advantageous Effects of Invention

(9) The present invention provides an exhaust purification apparatus capable of improving the stirring property of urea water and appropriately cooling an injection device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view illustrating an exhaust purification apparatus according to an embodiment of the present invention and an engine to which the exhaust purification apparatus is applied.
- (2) FIG. 2 is a perspective view illustrating an exhaust pipe and a through pipe in the exhaust purification apparatus according to the embodiment of the present invention.
- (3) FIG. 3 is a front view illustrating an exhaust pipe and a through pipe in the exhaust purification apparatus according to the embodiment of the present invention.
- (4) FIG. 4 is a front view illustrating an exhaust pipe and a through pipe in an exhaust purification apparatus according to an other example of the embodiment of the present invention.
- (5) FIG. 5 is a front view illustrating an exhaust pipe and a through pipe in an exhaust purification apparatus according to an other example of the embodiment of the present invention.
- (6) FIG. 6 is a front view illustrating an exhaust pipe and a through pipe in an exhaust purification apparatus according to an other embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

- (7) An exhaust purification apparatus **1** according to an embodiment of the present invention will be described with reference to the drawings. As illustrated in FIG. 1, the exhaust purification apparatus **1** is applied to an engine **100** and purifies exhaust gas discharged from the engine **100**. FIG. 1 illustrates an example in which the exhaust purification apparatus **1** is applied to the engine **100** that discharges exhaust gas in a vertical direction.
- (8) The engine **100** includes a plurality of cylinders each having a cylinder head or a cylinder block. The engine **100** includes an intake port (not illustrated) that takes in fuel gas, and an exhaust port **101** that discharges exhaust gas in a predetermined exhaust direction (for example, upward). An exhaust pipe **10** of the exhaust purification apparatus **1** is connected to the exhaust port **101**, and the engine **100** discharges exhaust gas to the exhaust pipe **10** of the exhaust purification apparatus **1** via the exhaust port **101**.
- (9) The exhaust purification apparatus **1** is constituted by a so-called selective catalytic reduction (SCR) system that purifies exhaust gas by reducing nitrogen oxides (NOx) in the exhaust gas. For example, as illustrated in FIG. 3, the exhaust purification apparatus **1** includes an exhaust pipe **10**, a catalyst reactor **11** (catalyst), a through pipe **12**, an injection device **13**, a reducing agent supply device **14**, and a cooling medium supply device **15**.
- (10) The main parts (the exhaust pipe **10**, the catalyst reactor **11**, the through pipe **12**, and the injection device **13**) of the exhaust purification apparatus **1** are provided with respect to the exhaust port **101** of the engine **100**, and are arranged, for example, above the exhaust port **101**. The reducing agent supply device **14** and the cooling medium supply device **15** may be provided separately from the main parts of the exhaust purification apparatus **1**. The reducing agent supply device **14** and the cooling medium supply device **15** may be integrally configured as one unit as illustrated in FIG. 1, or may be configured as separate bodies.
- (11) The exhaust pipe **10** allows exhaust gas discharged from the engine **100** to flow in a predetermined exhaust direction (for example, upward). The exhaust pipe **10** is coupled to an upper side of the exhaust port **101** of the engine **100**. The exhaust pipe **10** includes a straight pipe portion **10a** formed in a linear shape.
- (12) The catalyst reactor **11** brings exhaust gas mixed with a reducing agent such as urea water into contact with a catalyst and promotes a reduction reaction in which nitrogen oxides contained in the exhaust gas are selectively reduced to nitrogen and water, thereby purifying the exhaust gas. The catalyst reactor **11** is provided with a catalyst such as a NOx reduction catalyst and is connected to the exhaust pipe **10** on the downstream side in the exhaust direction. The catalyst reactor **11** introduces exhaust gas mixed with a reducing agent injected from the injection device **13** and flowing through the exhaust pipe **10**, and brings the exhaust gas into contact with the catalyst.

- (13) The through pipe **12** is provided in the exhaust pipe **10** on the upstream side of the catalyst reactor **11** (catalyst) in the exhaust direction in such a way as to pass through the center of the exhaust pipe **10** in the radial direction and penetrate the exhaust pipe **10** in the radial direction (direction orthogonal to the exhaust direction). The through pipe **12** may be provided on a straight pipe portion **10a** formed in a linear shape in the exhaust pipe **10**. The through pipe **12** is a cylindrical member of which both ends are opened, and the through pipe **12** is provided inside the exhaust pipe **10** by connecting the both ends of the through pipe **12** to openings provided on two opposing side surfaces of the exhaust pipe **10** by welding or the like. The exhaust gas flowing inside the exhaust pipe **10** flows on the outer peripheral surface side of the through pipe **12**.
- (14) The through pipe **12** has a substantially elliptical cross-sectional shape which is long in the exhaust direction, is formed in such a way that the cross-sectional shape is line-symmetrical with respect to the long axis thereof, and has the cross-sectional shape uniformly in the through direction. The through pipe **12** is formed to have a streamlined side surface with no corners.
- (15) The through pipe **12** includes a pair of side wall portions **20** extending in the exhaust direction, an upstream side portion **21** provided over the pair of side wall portions **20** on the upstream side in the exhaust direction, and a downstream side portion **22** provided over the pair of side wall portions **20** on the downstream side in the exhaust direction, and has a substantially elliptical cross-sectional shape. The through pipe **12** is configured in such a way such that the downstream side portion **22** is positioned inside both ends of the upstream side portion **21** in the width direction orthogonal to the through direction and the long axis of the cross-sectional shape.
- (16) The upstream side portion **21** of the through pipe **12** is formed to have, for example, a semicircular or semielliptical arc-shaped cross section which is convex toward the upstream side in the exhaust direction, and causes the exhaust gas flowing from the upstream side to flow to the outer peripheral surface side of the pair of side wall portions **20**.
- (17) The pair of side wall portions **20** of the through pipe **12** is formed to be curved in a streamlined shape in such a way that a width of the cross-sectional shape decreases toward the downstream side in the exhaust direction. For example, the pair of side wall portions **20** is formed to be curved with a curvature smaller than that of the upstream side portion **21**. For example, the side wall portion **20** is formed to have a linear portion **20a** extending continuously from an end part of the upstream side portion **21** in the exhaust direction and a curved portion **20b** curved continuously from a downstream end part of the linear portion **20a** to the center side in the width direction of the cross-sectional shape. The side wall portion **20** causes the exhaust gas flowing from the upstream side to flow along the outer peripheral surface, and creates a flow toward the center side in the width direction of the cross-sectional shape toward the downstream side.
- (18) The downstream side portion **22** of the through pipe **12** has a planar portion **22a** orthogonal to the exhaust direction and is formed continuously from downstream end parts of the pair of side wall portions **20**.
- (19) The through pipe **12** is formed by, for example, joining members having the same shape by welding or the like, and by providing a welding position in the upstream side portion **21** or the downstream side portion **22**, it is possible to form a side surface having a streamlined shape with no corners, without providing a weld overlay portion on the side surface.
- (20) With the above-described configuration of the through pipe **12**, the exhaust gas flowing on the outer peripheral surface side of the through pipe **12** is branched to both sides in the width direction by the upstream side portion **21**, then flows toward the center side in the width direction without stagnation along the pair of side wall portions **20** having a streamlined shape toward the center side in the width direction, and merges on the downstream side of the downstream side portion **22** without forming a dead water region such as a vortex retention region. In other words, the through pipe **12** is formed to have the pair of side wall portions **20** having a streamlined shape in such a way as not to form a dead water region on the downstream side of the downstream side portion **22**.
- (21) The injection device **13** is provided inside the through pipe **12** with only the tip having an

ejection hole exposed to the inside of the exhaust pipe **10**. For example, the injection device **13** is constituted by an injection nozzle, an injection valve, or the like. The injection device **13** is connected to the reducing agent supply device **14** and injects a reducing agent such as urea water for reducing exhaust gas supplied from the reducing agent supply device **14** into the exhaust pipe **10**. Note that the injection device **13** may be arranged in such a way as to inject the reducing agent to a position where the exhaust gases merge on the downstream side of the downstream side portion **22** of the through pipe **12**.

(22) The reducing agent supply device **14** is connected to a control device (not illustrated), and injects the reducing agent through the injection device **13** in response to an instruction signal from the control device. The injection device **13** is cooled by a cooling medium such as air flowing inside the through pipe **12**.

(23) The injection device **13** is provided on the downstream side in the exhaust direction between the pair of side wall portions **20** inside the through pipe **12**. For example, the injection device **13** is provided on a downstream side portion **22** of the through pipe **12**, to be specific, on a planar portion **22a** of the through pipe **12**. The injection device **13** is provided in such a way as to inject the reducing agent toward the downstream side in the exhaust direction, for example, toward the catalyst reactor **11** (catalyst) arranged on the downstream side. For example, one injection device **13** may be provided at the center in the through direction of the through pipe **12**, or a plurality of injection devices **13** may be provided at intervals in the through direction of the through pipe **12**. In the present embodiment, since the injection device **13** is arranged in the straight pipe portion **10a** of the exhaust pipe **10** and the catalyst reactor **11** is arranged at a downstream end of the straight pipe portion **10a**, the injection device **13** is configured to inject the reducing agent toward the catalyst reactor **11**, but the present invention is not limited to this embodiment. Alternatively, in an other example, the catalyst reactor **11** may be arranged in a bent pipe of the exhaust pipe **10** instead of the straight pipe portion **10a**. To be specific, a bent pipe may be connected to the downstream end of the straight pipe portion **10a**, and the catalyst reactor **11** may be connected to a downstream end of the bent pipe. In this case, the injection device **13** may be arranged in the straight pipe portion **10a**, or may be arranged in the bent pipe. Namely, the reducing agent may be injected toward the downstream side of the inside of the exhaust pipe **10** in the exhaust direction.

(24) The reducing agent supply device **14** is connected to the injection device **13** via a reducing agent supply pipe **16**, sucks up the reducing agent stored in a reducing agent tank **17** by using a pump or the like, and supplies the reducing agent to the injection device **13** via the reducing agent supply pipe **16**.

(25) The cooling medium supply device **15** is connected to the injection device **13** via a cooling medium supply pipe **18**, and supplies a cooling medium such as air supplied from an air supply section **19** to the injection device **13** via the cooling medium supply pipe **18**. For example, the cooling medium supply device **15** may introduce air cooled by an intercooler (not illustrated) and supply the air as a cooling medium. Alternatively, the cooling medium supply device **15** may supply cooling water as the cooling medium instead of air.

(26) As described above, according to the present invention, the exhaust purification apparatus **1** for purifying nitrogen oxides in exhaust gas includes the exhaust pipe **10** that causes exhaust gas discharged from the engine **100** to flow in a predetermined exhaust direction, the catalyst reactor **11** (catalyst) provided in the exhaust pipe **10** and reduces the nitrogen oxides in the exhaust gas, the through pipe **12** provided on the upstream side of the catalyst reactor **11** in the exhaust direction, in such a way as to penetrate the exhaust pipe **10** in a direction orthogonal to the exhaust direction, and the injection device **13** provided inside the through pipe **12** and injects a reducing agent for reducing exhaust gas into the exhaust pipe **10**.

(27) Thus, according to the exhaust purification apparatus **1** of the present invention, the injection device **13** that injects the reducing agent at the center position of the exhaust pipe **10** can be easily arranged and maintained by using the through pipe **12**. Further, by using the through pipe **12**, the

reducing agent can be injected at the center position of the exhaust pipe **10** without inserting most of the injection device **13** into the exhaust pipe **10**. Furthermore, by using the through pipe **12** with both ends penetrated, air permeability inside the through pipe **12** is improved, an influence of radiant heat from the wall surface of the through pipe **12** is suppressed, the inside of the through pipe **12** is suppressed from becoming a heat reservoir, cooling performance at the position of the injection device **13** can be improved, and thermal damage to the injection device **13** can be suppressed.

(28) Further, in the exhaust purification apparatus **1** of the present invention, the injection device **13** is provided inside the through pipe **12** on the downstream side in the exhaust direction in such a way as to inject the reducing agent toward the catalyst reactor **11**.

(29) Thus, since the reducing agent can be injected into the exhaust gas flowing toward the catalyst reactor **11**, stirring efficiency of the reducing agent can be improved, and generation of deposits can be suppressed.

(30) Further, in the exhaust purification apparatus **1** of the present invention, the through pipe **12** has the planar portion **22a** orthogonal to the exhaust direction on the downstream side in the exhaust direction, and the injection device **13** is provided on the planar portion **22a** of the through pipe **12**.

(31) As a result, accuracy of the injection direction of the injection device **13** can be improved, the stirring efficiency of the reducing agent can be improved, and generation of deposits can be suppressed.

(32) Further, in the exhaust purification apparatus **1** of the present invention, the through pipe **12** is provided in the straight pipe portion **10a** formed in a linear shape in the exhaust pipe **10**.

(33) Thus, the reducing agent can be injected into the exhaust gas traveling straight toward the catalyst reactor **11**, and therefore, the reducing agent can be prevented from adhering to the exhaust pipe **10** and the through pipe **12**.

(34) Further, in the exhaust purification apparatus **1** of the present invention, the through pipe **12** has the pair of side wall portions **20** extending in the exhaust direction and has a substantially elliptical cross-sectional shape elongated in the exhaust direction, and the pair of side wall portions **20** are formed to be curved in a streamlined shape, in such a way that the width of the cross-sectional shape decreases toward the downstream side in the exhaust direction.

(35) Thus, the exhaust gas flowing on the outer peripheral surface side of the pair of side wall portions **20** can be suppressed from being separated from the pair of side wall portions **20**, the exhaust gas can be caused to flow toward the center side in the width direction along the pair of side wall portions **20** without stagnation, and the exhaust gas can be caused to merge on the downstream side without forming a dead water region such as a vortex retention region. Therefore, regardless of penetration force of the injection device **13**, the reducing agent injected from the injection device **13** can be efficiently mixed with the exhaust gas and supplied to the catalyst reactor **11**.

(36) In addition, in the exhaust purification apparatus **1** of the present invention, the through pipe **12** has the upstream side portion **21** having an arc-shaped cross section on the upstream side of the pair of side wall portions **20** in the exhaust direction, and the pair of side wall portions **20** is formed to be curved with a curvature smaller than that of the upstream side portion **21**.

(37) Thus, it is possible to improve the performance of causing the exhaust gas to flow toward the center side in the width direction along the pair of side wall portions **20** without stagnation.

(38) Further, in the exhaust purification apparatus **1** of the present invention, the through pipe **12** is formed in such a way that the cross-sectional shape thereof is line symmetrical with respect to the central axis extending in the exhaust direction.

(39) Thus, the flow of the exhaust gas on the outer peripheral surface side of the through pipe **12** can be made symmetrical inside the exhaust pipe **10**, and the injected reducing agent can be prevented from flowing to the inner wall surface side of the exhaust pipe **10**.

(40) In the above-described embodiment, the example in which the pair of side wall portions **20** constituting the streamlined shape of the side surface of the through pipe **12** has the linear portion **20a** extending in the exhaust direction continuously from the end part of the upstream side portion **21** has been described, but the present invention is not limited to this example. In an other example, as illustrated in FIG. 4, the pair of side wall portions **20** may have a curved portion **20b** constituting a streamlined shape, which is formed continuously from the end part of the upstream side portion **21**, without having a linear portion **20a**.

(41) Further, in the above-described embodiment, the example in which the through pipe **12** has the downstream side portion **22** including the planar portion **22a** orthogonal to the exhaust direction has been described, but the present invention is not limited to this example. In an other example, as illustrated in FIG. 5, the through pipe **12** may have a pair of curved portions **20b** joined at the center in the width direction, without having the downstream side portion **22**. In this case, the injection device **13** may be provided at a joint between the pair of curved portions **20b**.

(42) Further, in an other embodiment, as illustrated in FIG. 6, the through pipe **12** may include a protrusion **20c** protruding outward in the width direction on at least one of the pair of side wall portions **20**. The protrusion **20c** acts as a vortex generator outside the side wall portion **20**, and after the exhaust gas is separated from the side wall portion **20** by the protrusion **20c**, the protrusion generates a vortex flow toward the downstream side in the exhaust direction and toward the center side in the width direction. The protrusion **20c** is provided, for example, in a transition portion where the side wall portion **20** transitions from the linear portion **20a** to the curved portion **20b**. In this case, the side wall portion **20** provided with the protrusion **20c** may not have a streamlined shape. One protrusion **20c** may be provided at the center in the through direction, or a plurality of protrusions **20c** may be provided at intervals in the through direction.

(43) According to such an other embodiment, by generating a vortex acting on the exhaust gas flowing on the outer peripheral surface side of the side wall portion **20**, the exhaust gas can be suppressed from being separated from the side wall portion **20**, the exhaust gas can be caused to flow toward the center side in the width direction on the downstream side of the through pipe **12**, and the exhaust gas can be caused to merge on the downstream side without forming a dead water region such as a vortex retention region. Therefore, regardless of penetration force of the injection device **13**, the reducing agent injected from the injection device **13** can be efficiently mixed with the exhaust gas and supplied to the catalyst reactor **11**.

(44) Note that the present invention can be properly modified within a range which does not contradict to a gist or a concept of the present invention that can be read throughout the claims and the specification, and an exhaust purification apparatus involving such modifications is also included in the technical concept of the present invention.

Supplementary Notes

(45) Hereinafter, the summary of the invention extracted from the above-described embodiments will be supplementarily noted. Configurations and processing functions to be described in the following supplementary notes may be selected and combined as appropriate.

(46) Supplementary Note 1

(47) An exhaust purification apparatus that purifies nitrogen oxides in exhaust gas, the exhaust purification apparatus including: an exhaust pipe that causes the exhaust gas discharged from an engine to flow; a catalyst that is provided in the exhaust pipe and that reduces the nitrogen oxides in the exhaust gas; a through pipe that is provided in such a way as to penetrate the exhaust pipe; and an injection device that is provided inside the through pipe and that injects a reducing agent that reduces the exhaust gas into the exhaust pipe.

Supplementary Note 2

(48) The exhaust purification apparatus according to supplementary note 1, wherein the injection device is provided in such a way as to inject the reducing agent to a downstream side in the exhaust direction.

(49) Supplementary Note 3

(50) The exhaust purification apparatus according to supplementary note 1 or 2, wherein the through pipe has a planar portion on a downstream side in the exhaust direction, and the injection device is provided on the planar portion.

Supplementary Note 4

(51) The exhaust purification apparatus according to any one of supplementary notes 1 to 3, wherein the through pipe is provided in a straight pipe portion formed in a linear shape in the exhaust pipe.

(52) Supplementary Note 5

(53) The exhaust purification apparatus according to any one of supplementary notes 1 to 4, wherein the through pipe includes a pair of side wall portions extending in the exhaust direction and has a substantially elliptical cross-sectional shape elongated in the exhaust direction, and the pair of side wall portions are formed in such a way that a width of the cross-sectional shape decreases toward a downstream side in the exhaust direction.

Supplementary Note 6

(54) The exhaust purification apparatus according to supplementary note 5, wherein the through pipe includes an upstream side portion having an arc-shaped cross section on an upstream side of the pair of side wall portions, and the pair of side wall portions are formed to be curved with a curvature smaller than that of the upstream side portion.

Supplementary Note 7

(55) The exhaust purification apparatus according to any one of supplementary notes 1 to 6, wherein the through pipe is formed in such a way that a cross-sectional shape thereof is line-symmetric with respect to a central axis extending in an exhaust direction.

REFERENCE SIGNS LIST

(56) **1** exhaust purification apparatus **10** exhaust pipe **10a** straight pipe portion **11** catalyst reactor (catalyst) **12** through pipe **13** injection device **14** reducing agent supply device **15** cooling medium supply device **16** reducing agent supply pipe **17** reducing agent tank **18** cooling medium supply pipe **19** air supply section **20** side wall portion **20a** linear portion **20b** curved portion **20c** protrusion **21** upstream side portion **22** downstream side portion **100** engine **101** exhaust port

Claims

1. An exhaust purification apparatus that purifies nitrogen oxides in exhaust gas, the exhaust purification apparatus comprising: an exhaust pipe that causes the exhaust gas discharged from an engine to flow; a catalyst that is provided in the exhaust pipe and that reduces the nitrogen oxides in the exhaust gas; a through pipe, having both ends open, that is provided in such a way to pass through the exhaust pipe; and an injection device that is provided inside the through pipe and that injects a reducing agent, that reduces the exhaust gas, into the exhaust pipe, wherein the through pipe has a planar portion on a downstream side in an exhaust direction.
2. The exhaust purification apparatus according to claim 1, wherein the injection device is provided in such a way as to inject the reducing agent to a downstream side in an exhaust direction.
3. The exhaust purification apparatus according to claim 1, wherein the injection device is provided on the planar portion.
4. The exhaust purification apparatus according to claim 1, wherein the exhaust pipe has a straight pipe portion formed in a linear shape, the through pipe is provided in the straight pipe portion.
5. The exhaust purification apparatus according to claim 1, wherein the through pipe has a pair of side wall portions extending in an exhaust direction, and has a substantially elliptical cross-sectional shape elongated in the exhaust direction, and the pair of side wall portions is formed in such a way that a width of the cross-sectional shape decreases toward a downstream side in the exhaust direction.

6. The exhaust purification apparatus according to claim 5, wherein the through pipe has an upstream side portion having an arc-shaped cross section on an upstream side of the pair of side wall portions, and the pair of side wall portions is formed to be curved with a less tight curvature than that of the upstream side portion.
7. The exhaust purification apparatus according to claim 1, wherein the through pipe is formed in such a way that a cross-sectional shape thereof is line-symmetrical with respect to a central axis extending in an exhaust direction.
8. The exhaust purification apparatus according to claim 1, the injection device injects the reducing agent into the exhaust pipe downstream of the through pipe.
9. An exhaust purification apparatus that purifies nitrogen oxides in exhaust gas, the exhaust purification apparatus comprising: an exhaust pipe that causes the exhaust gas discharged from an engine to flow; a catalyst that is provided in the exhaust pipe and that reduces the nitrogen oxides in the exhaust gas; a through pipe, having both ends open, that is provided in such a way to pass through the exhaust pipe; and an injection device that is provided inside the through pipe and that injects a reducing agent, that reduces the exhaust gas, into the exhaust pipe, wherein the through pipe has an upstream side portion having an arc-shaped cross section on an upstream side.
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